RECAPITULARE ȘIRURI

- **Studiați monotonia următoarelor șiruri:**

1. $x_n = \dfrac{2n^2}{n+1},\ n \in \mathbb{N}$

2. $x_n = n \cdot \left(\dfrac{1}{3}\right)^n,\ n \in \mathbb{N}^*$

3. $x_n = \dfrac{1}{n+1} + \dfrac{1}{n+2} + \ldots + \dfrac{1}{2n},\ n \in \mathbb{N}^*$

4. $x_n = \dfrac{5 \cdot 14 \cdot \ldots \cdot \left(2n^2 - n - 1\right)}{9 \cdot 20 \cdot \ldots \cdot \left(2n^2 + n - 1\right)},\ n \geq 2$

5. $x_n = \dfrac{(-1)^n \cdot 3^n}{n!},\ n \in \mathbb{N}^*$

6. $x_n = 2 - (-1)^n \cdot \dfrac{1}{n},\ n \in \mathbb{N}^*$

- **Să se calculeze limita următoarelor șiruri:**

7. $\lim\limits_{n\to\infty}\left(n + \sqrt{n + \sqrt{n}} - \sqrt{n}\right)$

8. $\lim\limits_{n\to\infty}\left(\sqrt[3]{n^3 + 2n^2 + 1} - \sqrt[3]{n^3 - 3}\right)$

9. $\lim\limits_{n\to\infty} n \cdot \left(\sqrt{n^2 + 1} - n\right)$

10. $\lim\limits_{n\to\infty}\left(\dfrac{1^2 + 2^2 + \ldots + n^2}{n^2} - \dfrac{n}{3}\right)$

11. $\lim\limits_{n\to\infty}\left(\sqrt{n + \sqrt{n}} - \sqrt{n - \sqrt{n}}\right)$

12. $\lim\limits_{n\to\infty} \dfrac{\sum\limits_{k=1}^{n} k(k+3)}{C_{n+3}^{n}}$

13. $\lim\limits_{n\to\infty}\left(\sqrt{n + \sqrt{n+1}} - \sqrt{n+1}\right)$

14. $\lim\limits_{n\to\infty} \dfrac{1}{n^2}\left(3\sin n + 6\cos n\right)$

15. $\lim\limits_{n\to\infty} n \cdot \left(\sqrt{\dfrac{n+1}{n+2}} - 1\right)$

16. $\lim\limits_{n\to\infty}\left(2n - \sqrt[3]{8n^3 + 4n^2 - n}\right)$

17. $\lim\limits_{n\to\infty}\left(n - \sqrt[3]{n^3 - 1}\right)$

18. $\lim\limits_{n\to\infty} \dfrac{\cos n}{n}$

19. $\lim\limits_{n\to\infty} \dfrac{3n^2 - \sqrt{2n^3 + 5n}}{2n^2 + 3n + 1}$

20. $\lim\limits_{n\to\infty} \dfrac{\sqrt{n^2 + 1} + \sqrt{n}}{\sqrt[4]{n^3 + n} - n}$

21. $\lim\limits_{n\to\infty} \sqrt{\dfrac{2 \cdot 5^n + 3 \cdot 6^n}{5 \cdot 3^n + 6^{n+1}}}$

22. $\lim\limits_{n\to\infty}\left(1 + \dfrac{1}{\sqrt{n+1}}\right) \cdot \left(1 + \dfrac{1}{\sqrt{n}}\right)^{2\sqrt{n+1}}$

23. $\lim\limits_{n\to\infty}\left(\dfrac{5n+1}{5n}\right)^{6n-4} \cdot \left(\dfrac{n+1}{n}\right)^{5n+3}$

24. $\lim\limits_{n\to\infty}\left(\dfrac{2n+3}{2n+1}\right)^{n+1}$

25. $\lim\limits_{n\to\infty}\left(\dfrac{n!}{\left(1+1^2\right)\left(1+2^2\right)..\left(1+n^2\right)}\right)$

26. $\lim\limits_{n\to\infty}\left(\dfrac{7^n + 4^n}{2 \cdot 7^n + 5^n}\right)^{\frac{n^2}{4n^2-3}}$

27. $\lim\limits_{n\to\infty}\left(\dfrac{4^n + 3}{4^n + 1}\right)^{2^{2n+1}}$

28. $\lim\limits_{n\to\infty}\left(\sqrt{n^2 + 2n + 3} - n\right)^{n+\sqrt{n}}$

29. $\lim\limits_{n\to\infty} \dfrac{1}{n}\left(1 + \dfrac{1}{\sqrt{2}} + \ldots + \dfrac{1}{\sqrt{n}}\right)$

30. $\lim\limits_{n\to\infty}\left(\dfrac{n^n}{(n!)^2}\right)$

31. $\lim\limits_{n\to\infty} \dfrac{1}{n}\left(\ln 2 + \ln 3 + \ldots + \ln n\right)$

32. $\lim\limits_{n\to\infty} \dfrac{1 + \sqrt{2} + \ldots + \sqrt{n}}{n\sqrt{n}}$

33. $\lim\limits_{n\to\infty} \sqrt[n]{\dfrac{(n!)^2}{(2n)! \cdot 8^n}}$

34. $\lim\limits_{n\to\infty} \sqrt[n]{\dfrac{(n+1)(n+2)...(n+n)}{n}}$

35. $\lim\limits_{n\to\infty}\left(\dfrac{C_n^2}{2^n}\right)$

36. $\lim\limits_{n\to\infty} \sqrt[n]{\ln(n!)}$

- **Stabiliți natura următoarelor șiruri** *(natura unui șir reprezintă calitatea lui de a fi convergent sau divergent)*:

37. $x_n = \frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \ldots + \frac{1}{n\cdot(n+1)}$, $n \in \mathbb{N}^*$

38. $x_n = \frac{1\cdot 3\cdot 5\cdot \ldots \cdot (2n-1)}{2\cdot 4\cdot 6\cdot \ldots \cdot 2n}$, $n \in \mathbb{N}^*$

39. $x_n = \frac{1}{\sqrt{n^2+1}} + \frac{1}{\sqrt{n^2+2}} + \ldots + \frac{1}{\sqrt{n^2+n}}$, $n \in \mathbb{N}^*$

40. $x_n = \frac{1}{n}\sum_{k=1}^{n} \cos\left(\frac{\pi}{n+k}\right)$, $n \in \mathbb{N}^*$

41. $x_n = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \ldots + \frac{1}{n^2}$, $n \in \mathbb{N}^*$ (șirul lui Euler, $\lim\limits_{n\to\infty} x_n = \frac{\pi^2}{6}$)

42. $x_n = \frac{n}{n+1} + \left(1 + \frac{1}{n}\right)^n \cdot i$ (șir complex)

43. $x_n = \frac{n}{n+1} + (-1)^n \cdot i$ (șir complex)